



Climate Change and Allergies: How Warming Temperatures Amplify Pollen Seasons

Abstract

Warmer temperatures and rising carbon dioxide from burning fossil fuels are making pollen seasons longer and more intense. This lesson, based on a Harvard Chan C-CHANGE guide for families, explains how climate change is worsening hay fever and allergy-triggered asthma in children, what the research shows about pollen trends, and practical steps to reduce exposure.

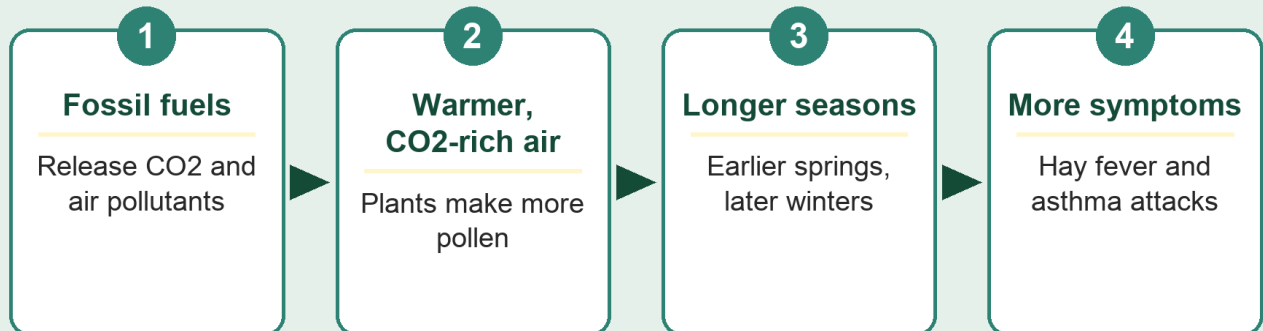
Introduction

Hay fever, also called allergic rhinitis, is one of the most common allergies in children, affecting 9% of kids in the United States. When a child with hay fever breathes in allergens such as pollen, their body overreacts, causing sneezing, coughing, itchy or watery eyes, and a stuffy or runny nose. Hay fever can also make asthma worse. For many families, allergy season is an annual challenge. But climate change is making that season longer and stronger. Carbon pollution and warmer temperatures cause plants to produce more pollen over longer growing seasons, and fossil fuel air pollution can make allergy symptoms even worse.

How Do Scientists Study This?

Scientists study the link between climate and allergies in several ways. Networks of pollen-collecting stations measure how much pollen is in the air each day and when the season begins, peaks, and ends. Comparing these records across decades reveals long-term trends. Researchers also run experiments showing that many allergenic plants produce more pollen when grown in air with more carbon dioxide. Using climate projections, scientists then estimate future pollen levels and timing. Finally, reports such as a 2023 U.S. Environmental Protection Agency (EPA) analysis connect projected pollen increases to health outcomes, such as children's asthma-related emergency department visits.

How climate change worsens allergies



Count pollen

Stations across the U.S. and Canada collect and measure airborne pollen every season.

Compare decades

Long-term records show when seasons start, peak, and end, and how that is changing.

Project health effects

Climate projections estimate future pollen levels and their effects on children's health.

What Did They Discover?

Pollen counts in the United States have been increasing for more than 20 years, at least partly because of higher temperatures and carbon dioxide. A study of 60 pollen stations in the U.S. and Canada found that pollen season is now 20 days longer on average than in 1990. Scientists predict average pollen counts in 2040 will be more than double those in 2000. In 2000, pollen production began on April 14 and peaked on May 1; by 2040, pollen may peak as early as April 8. The 2023 EPA report projected that pollen increases from 2°C to 4°C of warming would raise children's asthma-related emergency department visits by 17% to 30% each year.

Pollen by the numbers

9%

of U.S. kids have hay fever

+20 days

longer pollen season than in 1990

2×

pollen counts projected for 2040 versus 2000

Cutting fossil fuel use has a double benefit: it slows climate change and reduces allergy symptoms and asthma attacks triggered by pollen.

Takeaways

Longer seasons

Pollen season is 20 days longer than in 1990 and starting earlier.

More pollen

Pollen counts may double between 2000 and 2040 as warming continues.

Protect and prevent

Checking pollen levels and cutting emissions both protect children's health.

Key Terms

Allergen

A substance, such as pollen, that causes an allergic reaction in some people.

Hay fever (allergic rhinitis)

An allergy to airborne substances like pollen that causes sneezing, congestion, and itchy eyes.

Pollen count

A measure of how many pollen grains are in a set volume of air, used to forecast allergy risk.

Growing season

The part of the year when plants actively grow, which is lengthening as springs come earlier and winters later.

Carbon dioxide fertilization

The effect of extra carbon dioxide in the air causing some plants to grow more and produce more pollen.

Match-Up: draw a line from each term to its clue

A Allergen

B Hay fever (allergic rhinitis)

C Pollen count

D Growing season

E Carbon dioxide fertilization

1 The part of the year when plants actively grow, which is lengthening...

2 A measure of how many pollen grains are in a set volume...

3 An allergy to airborne substances like pollen that causes sneezing, congestion, and...

4 The effect of extra carbon dioxide in the air causing some plants...

5 A substance that causes an allergic reaction in some people

Answer key: A-5 · B-3 · C-2 · D-1 · E-4

Activity

Pollen Season Timeline

Students build a timeline comparing pollen seasons in 2000 and projected for 2040, then design an allergy action plan.

Materials: Calendar template for March through June; Colored pencils; Access to a pollen forecast website (optional)

1. On the calendar, mark the 2000 pollen season start (April 14) and peak (May 1) in one color.
2. Mark the projected 2040 peak (April 8) in another color. How many days earlier is it?
3. Add a note showing that the season is now about 20 days longer than in 1990.
4. Using the lesson, list four steps a family can take on high-pollen days.
5. If possible, check today's local pollen forecast and decide which steps you would take.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate, Kids, and Health: Allergies.
<https://hsph.harvard.edu/research/climate-health-c-change/climate-kids-and-health/allergies/>